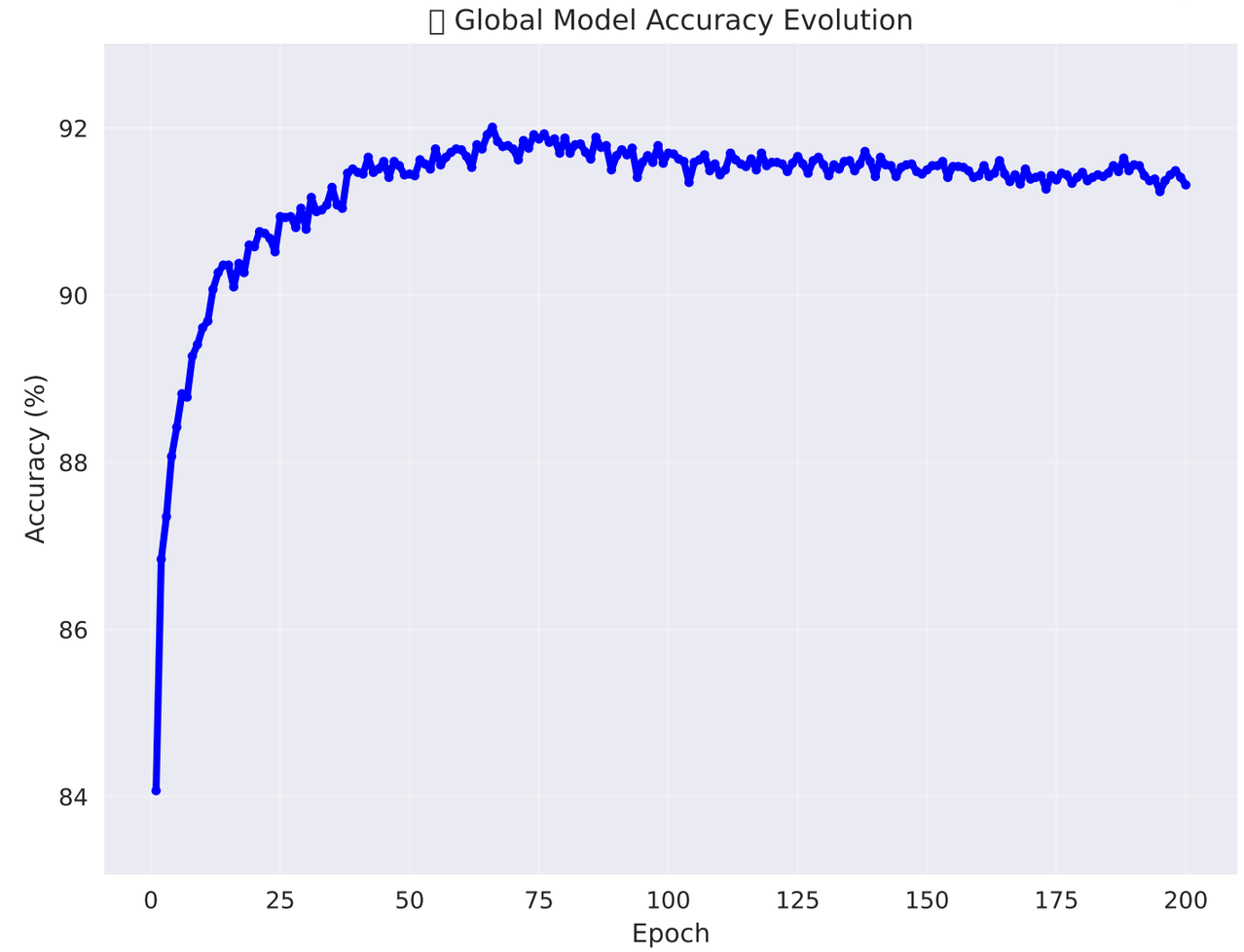
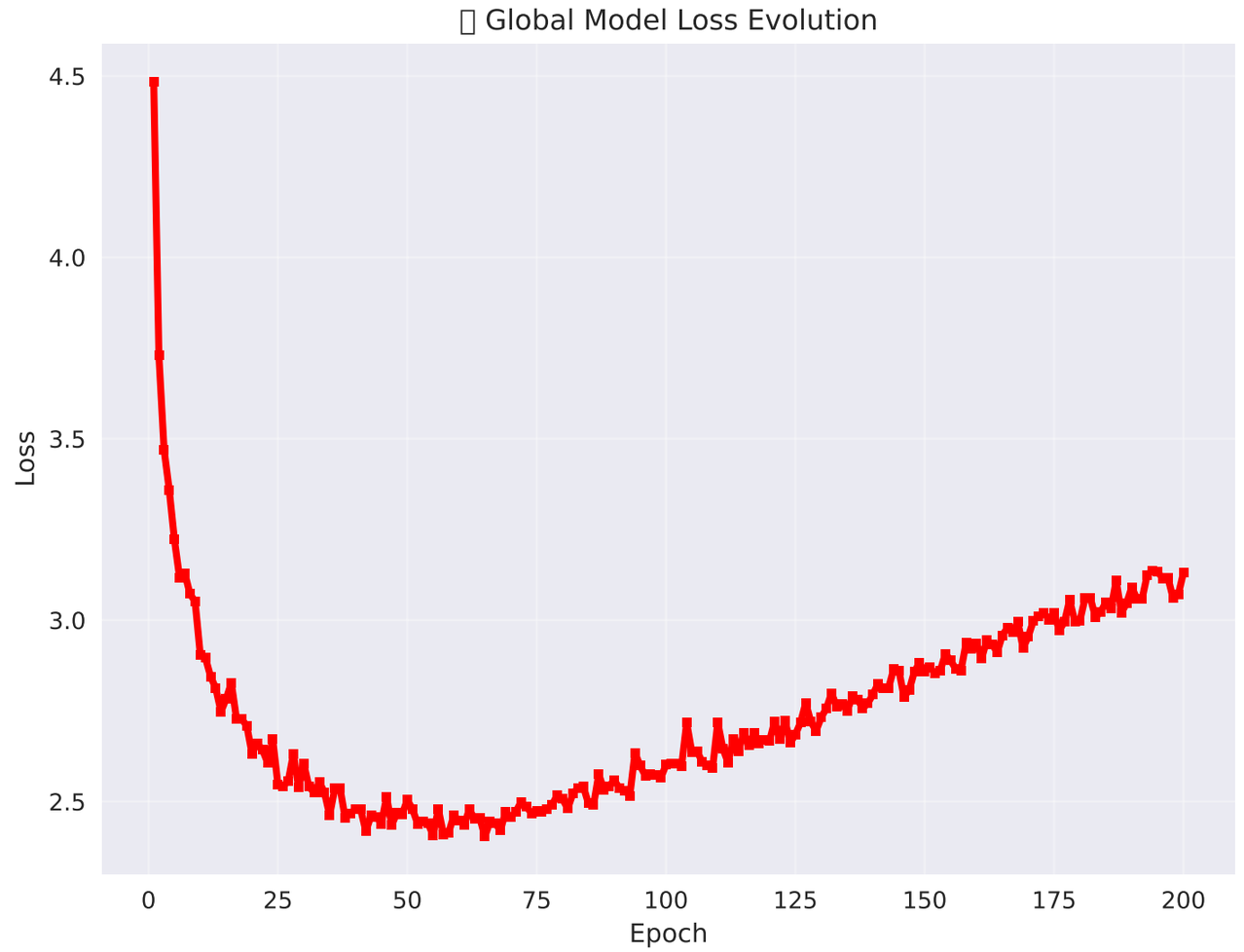
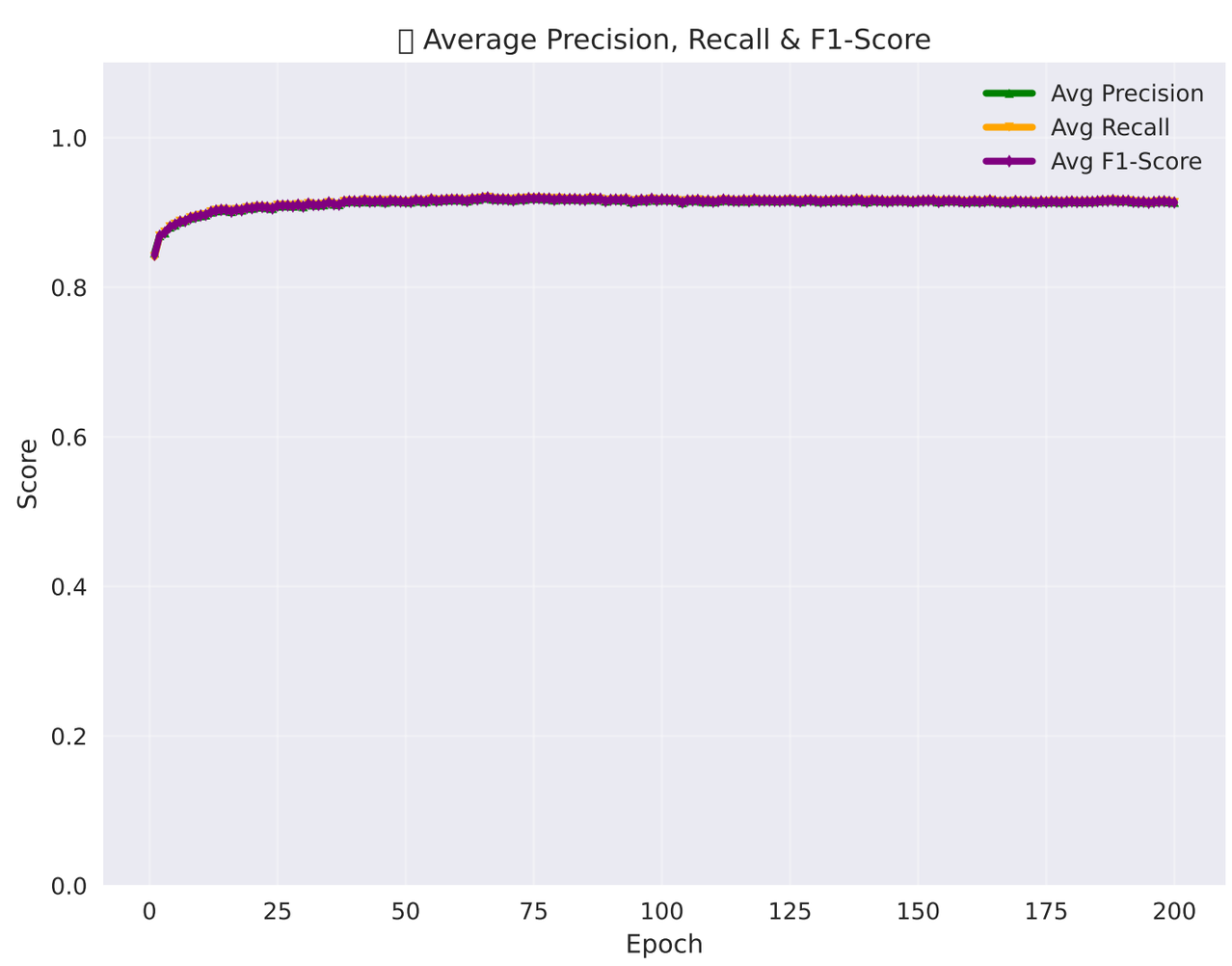
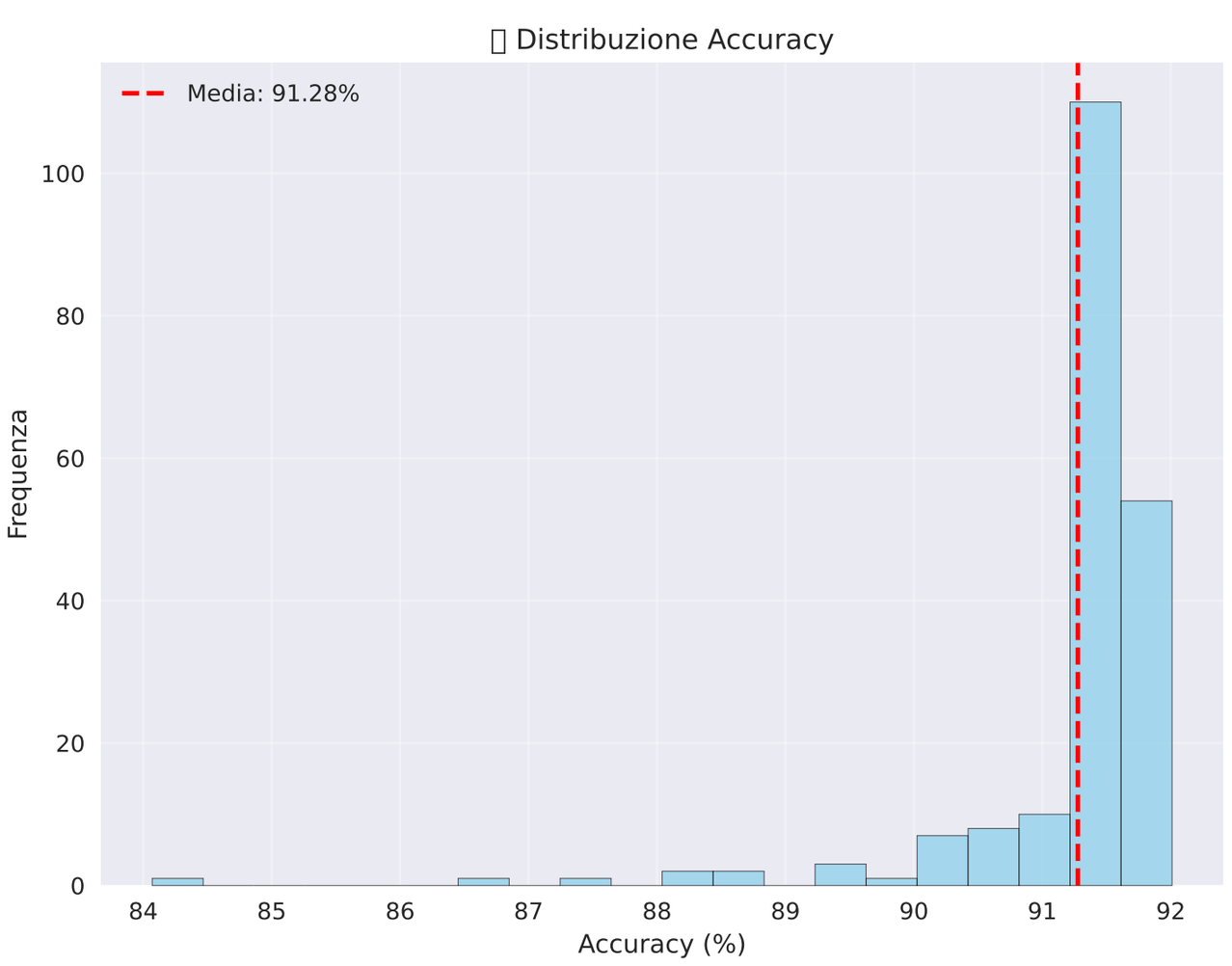
Federated Learning - Main Metrics Overview

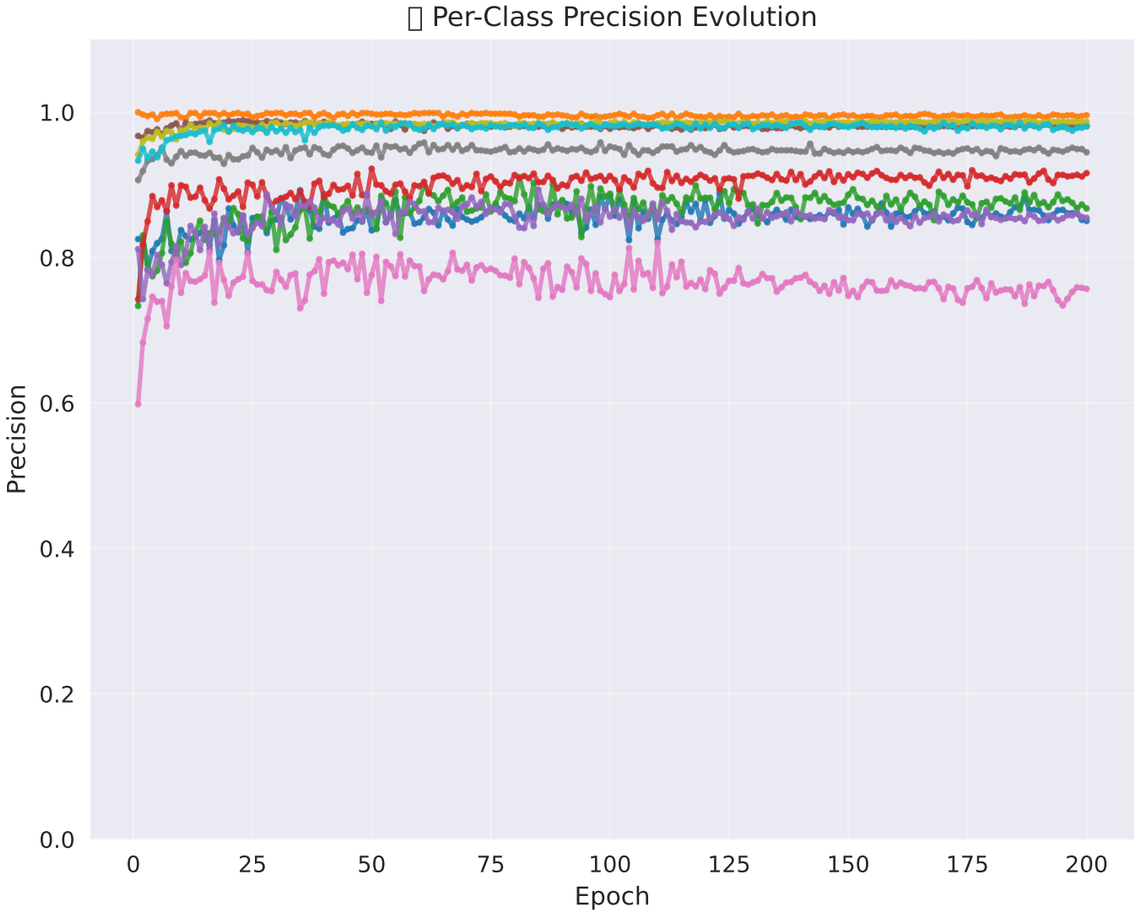


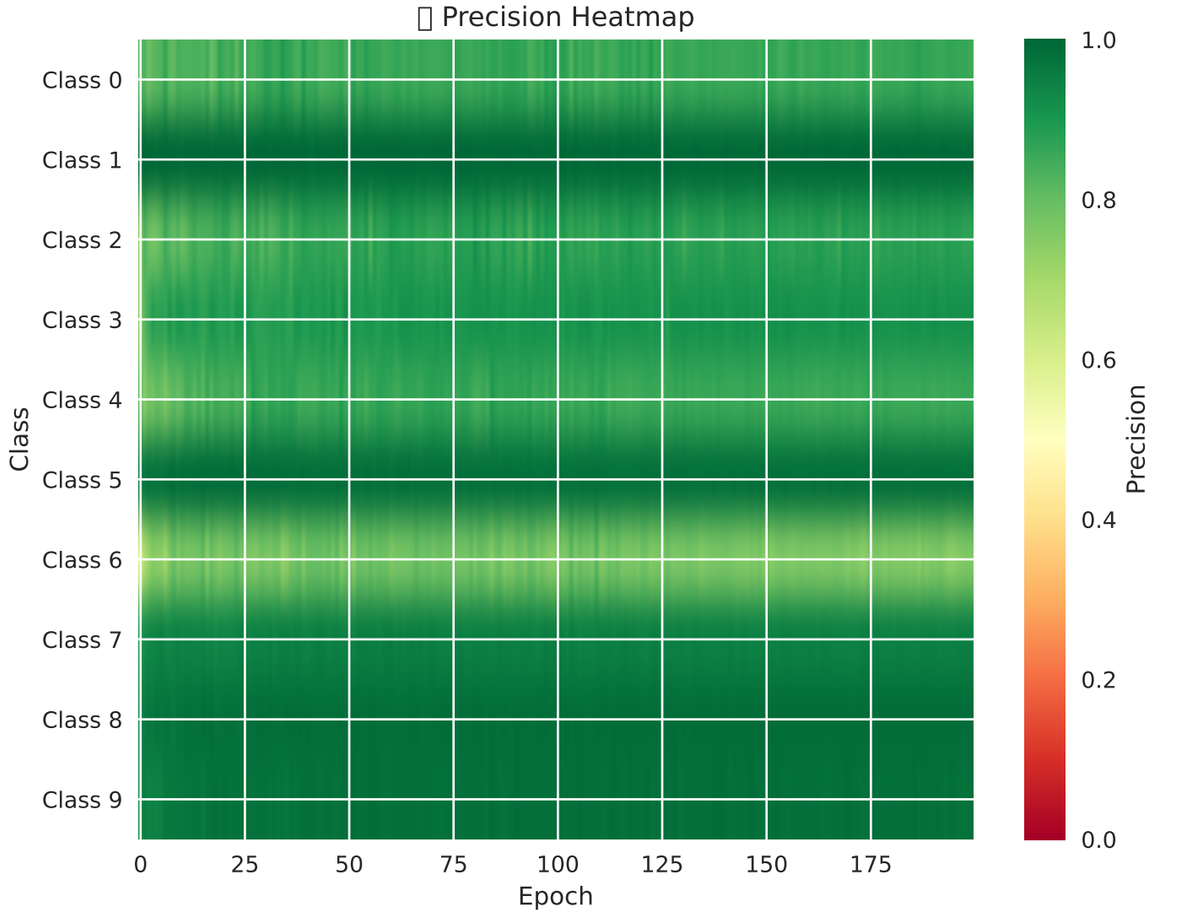


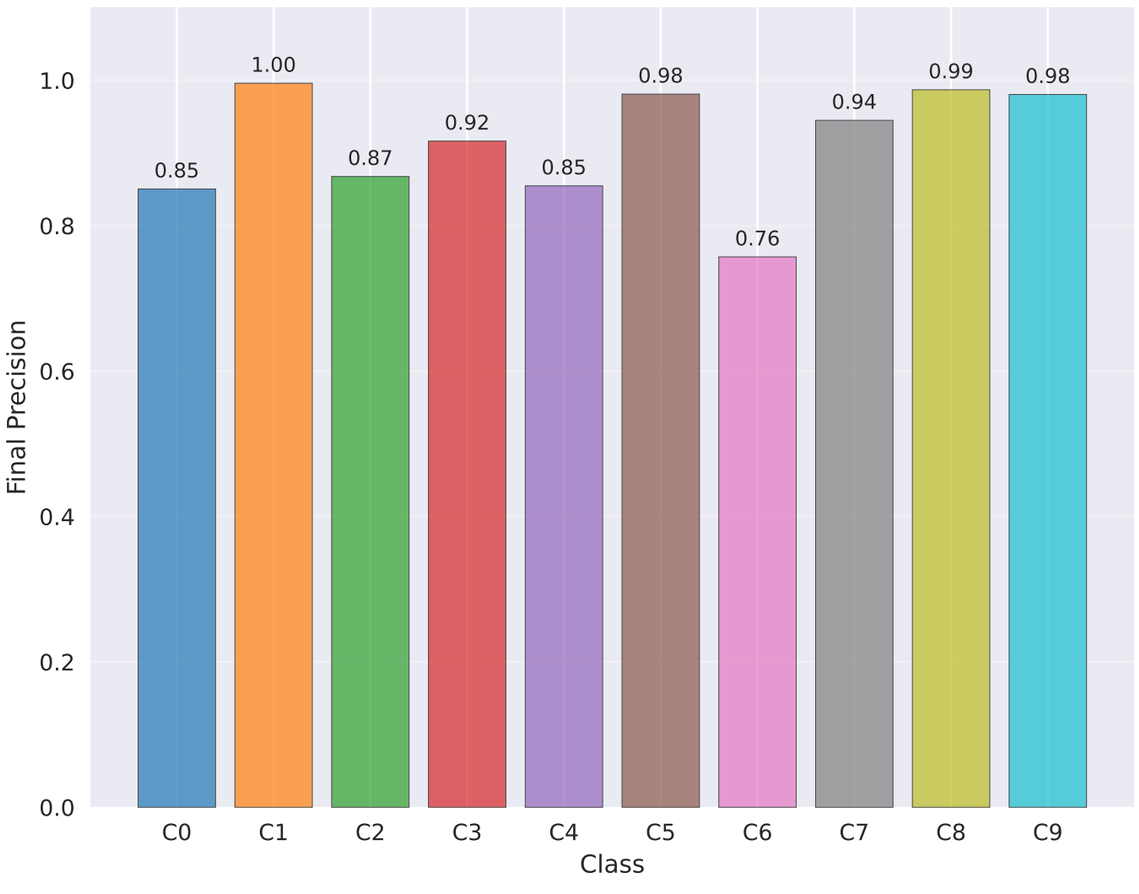


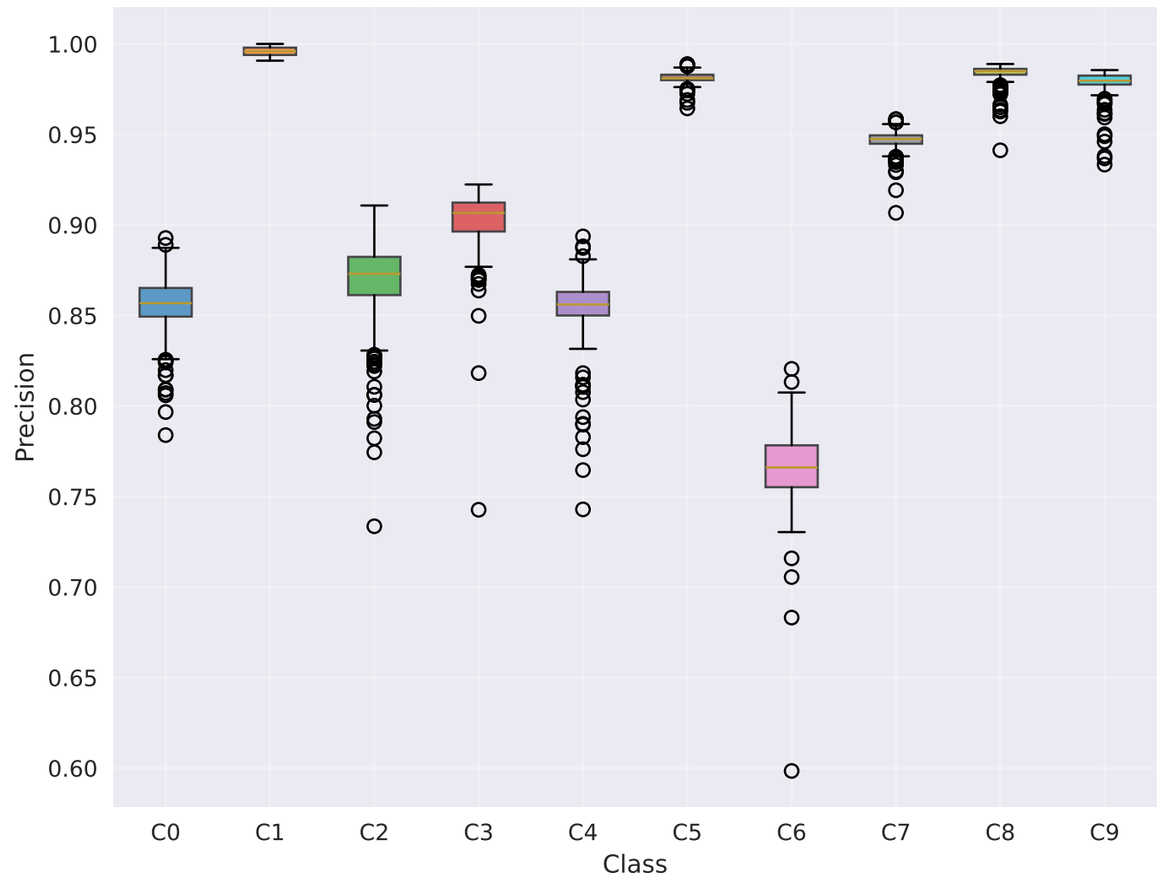


Per-Class Precision Analysis



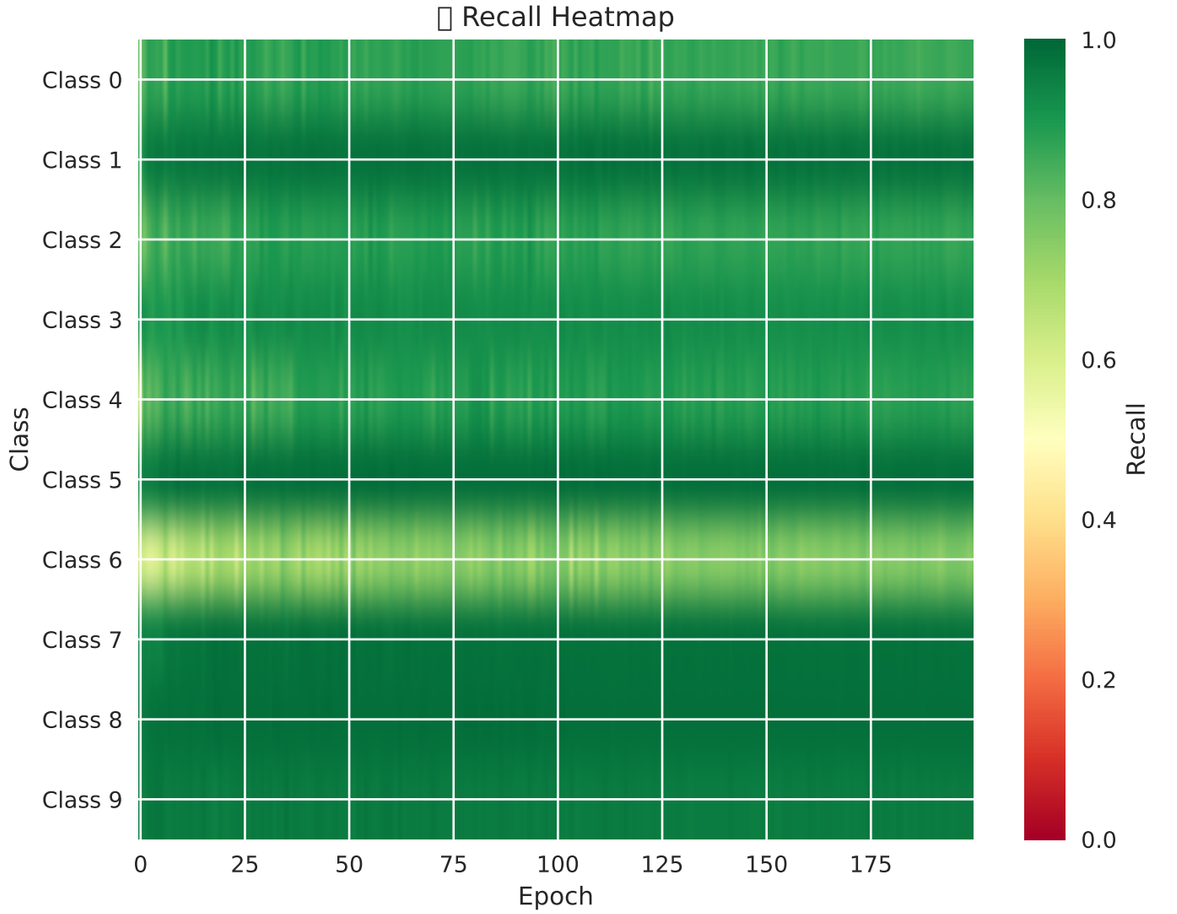


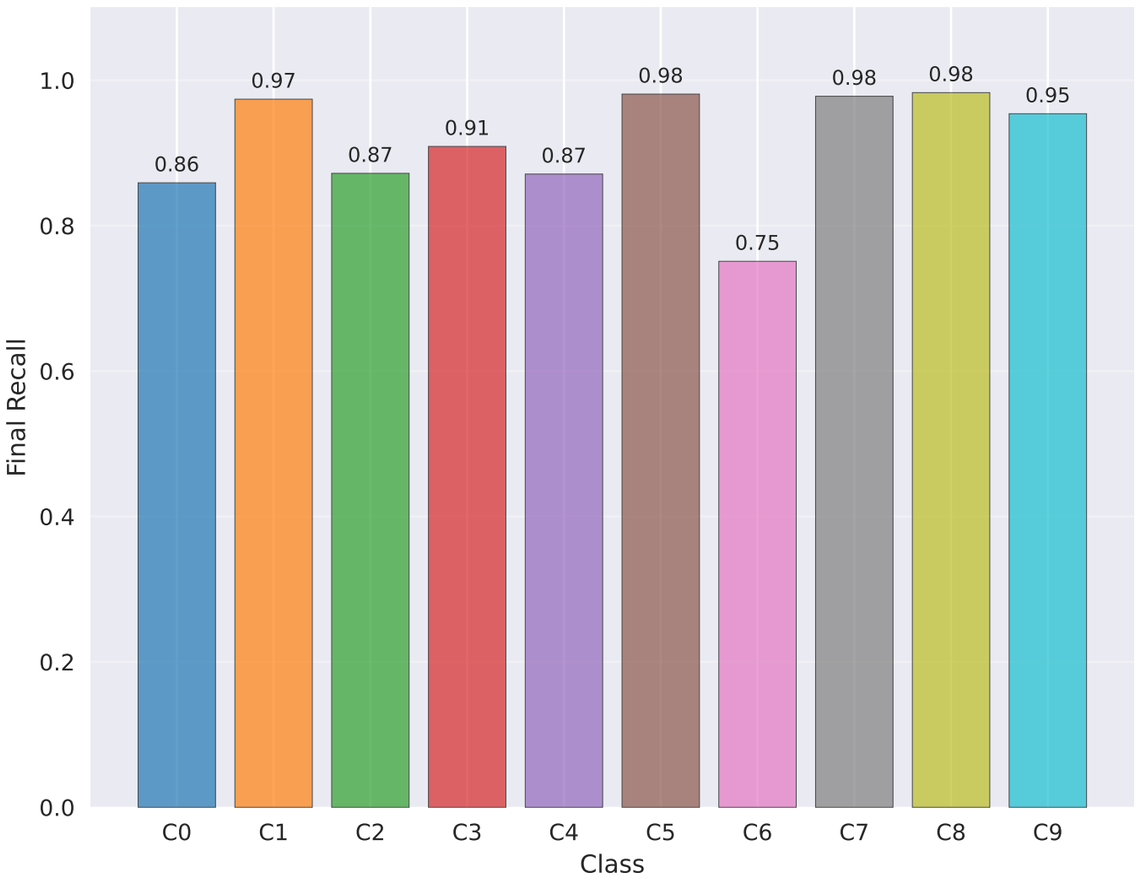


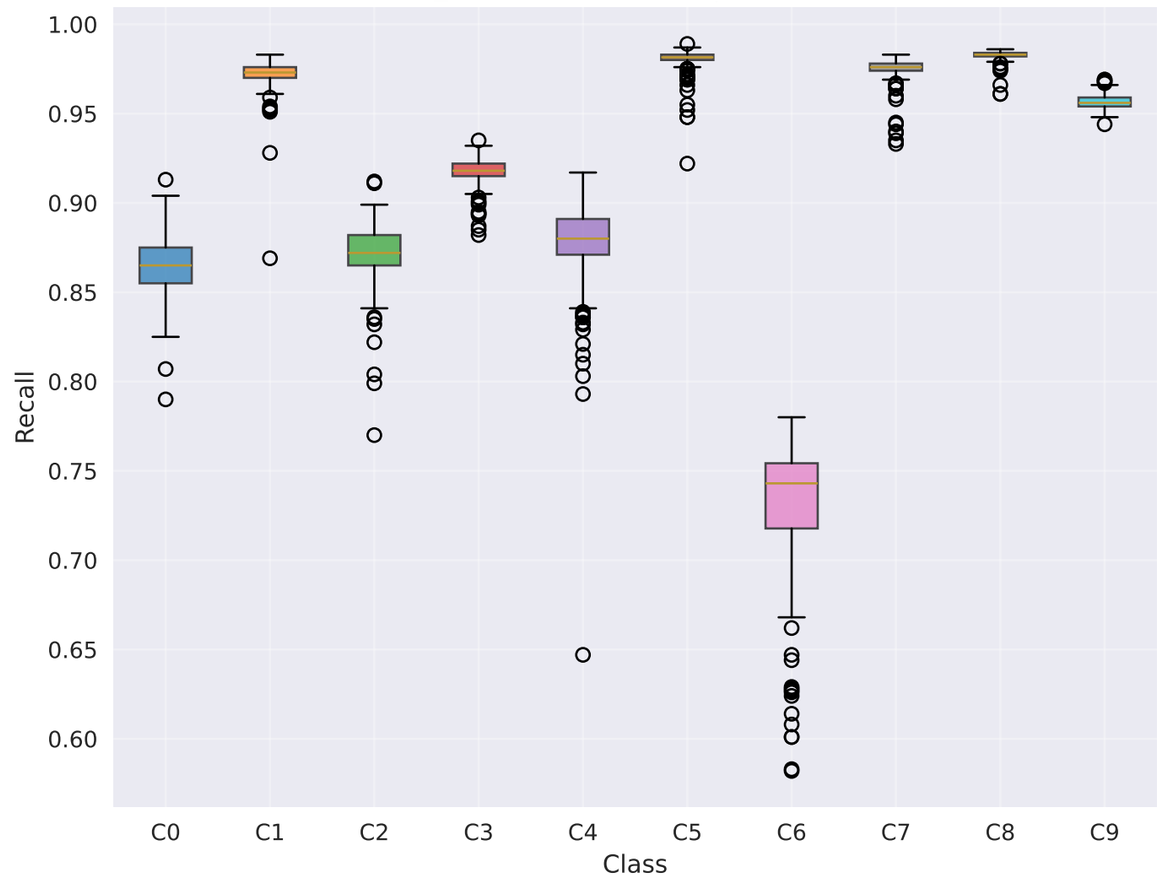


Per-Class Recall Analysis

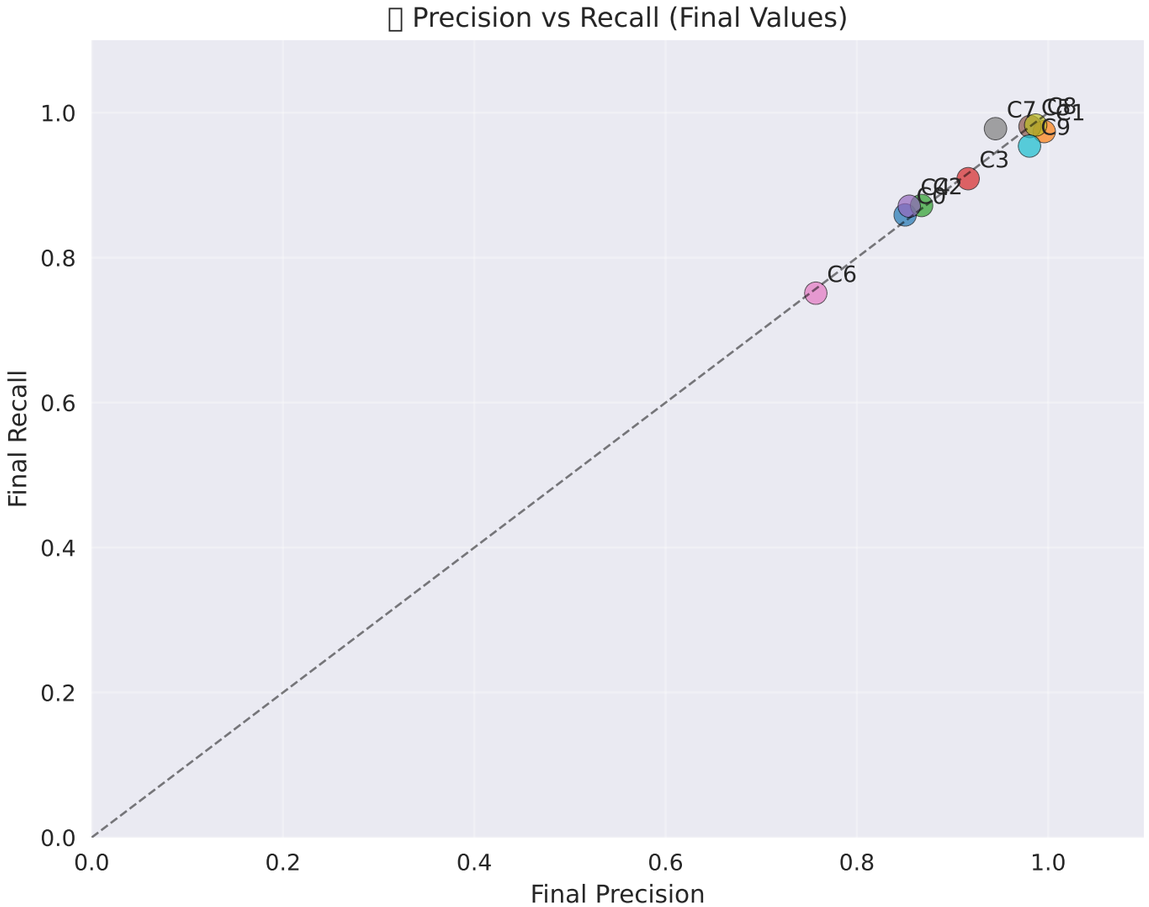


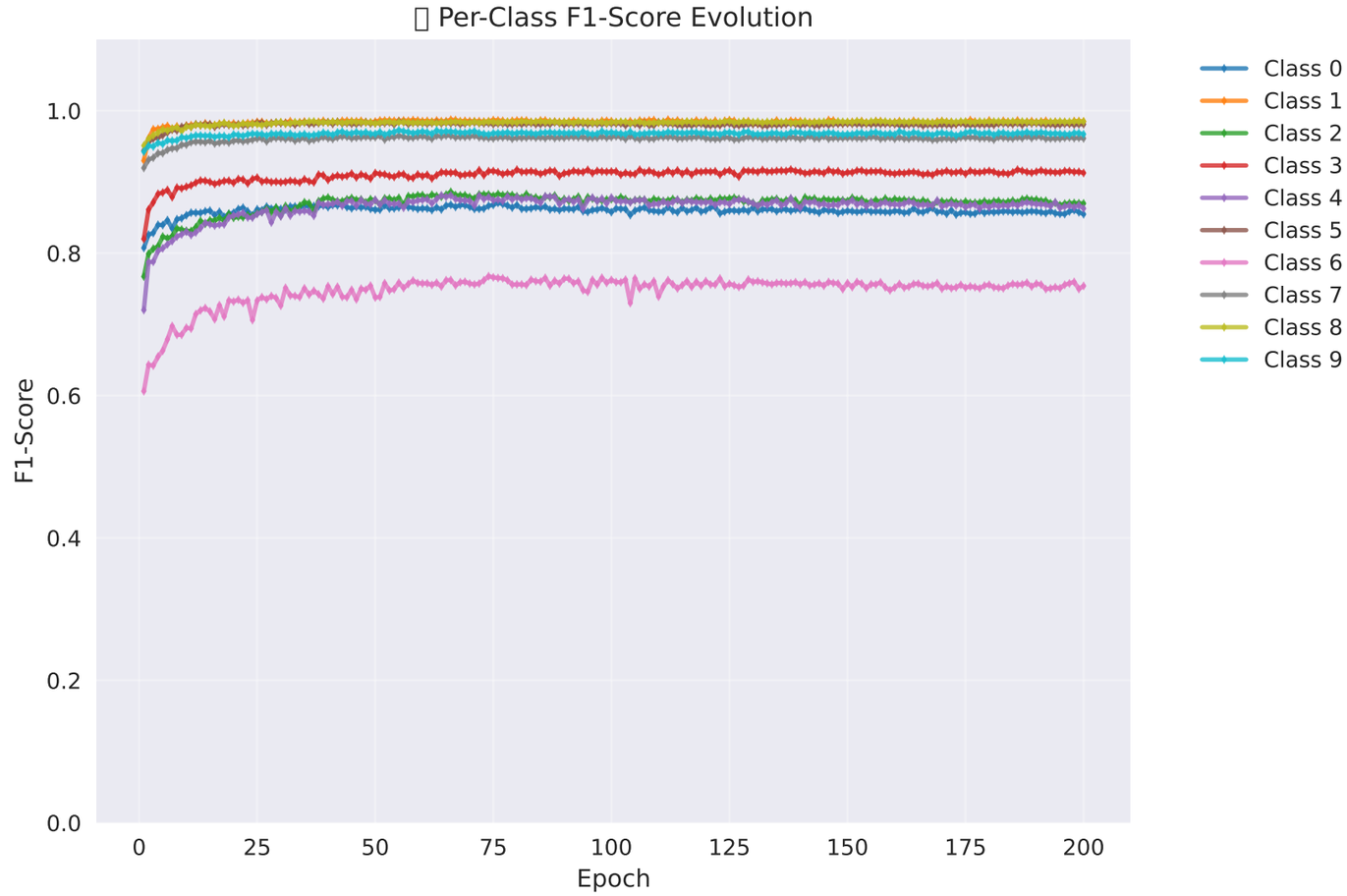


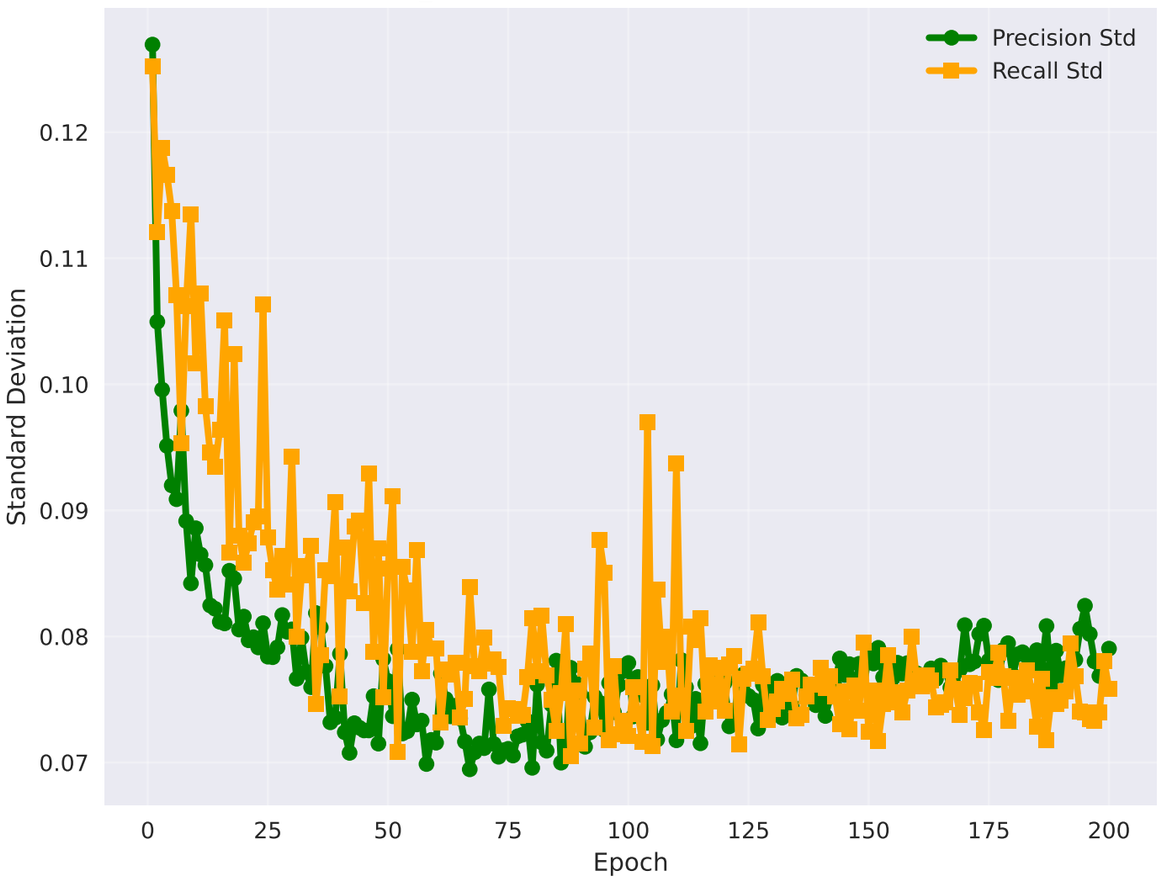


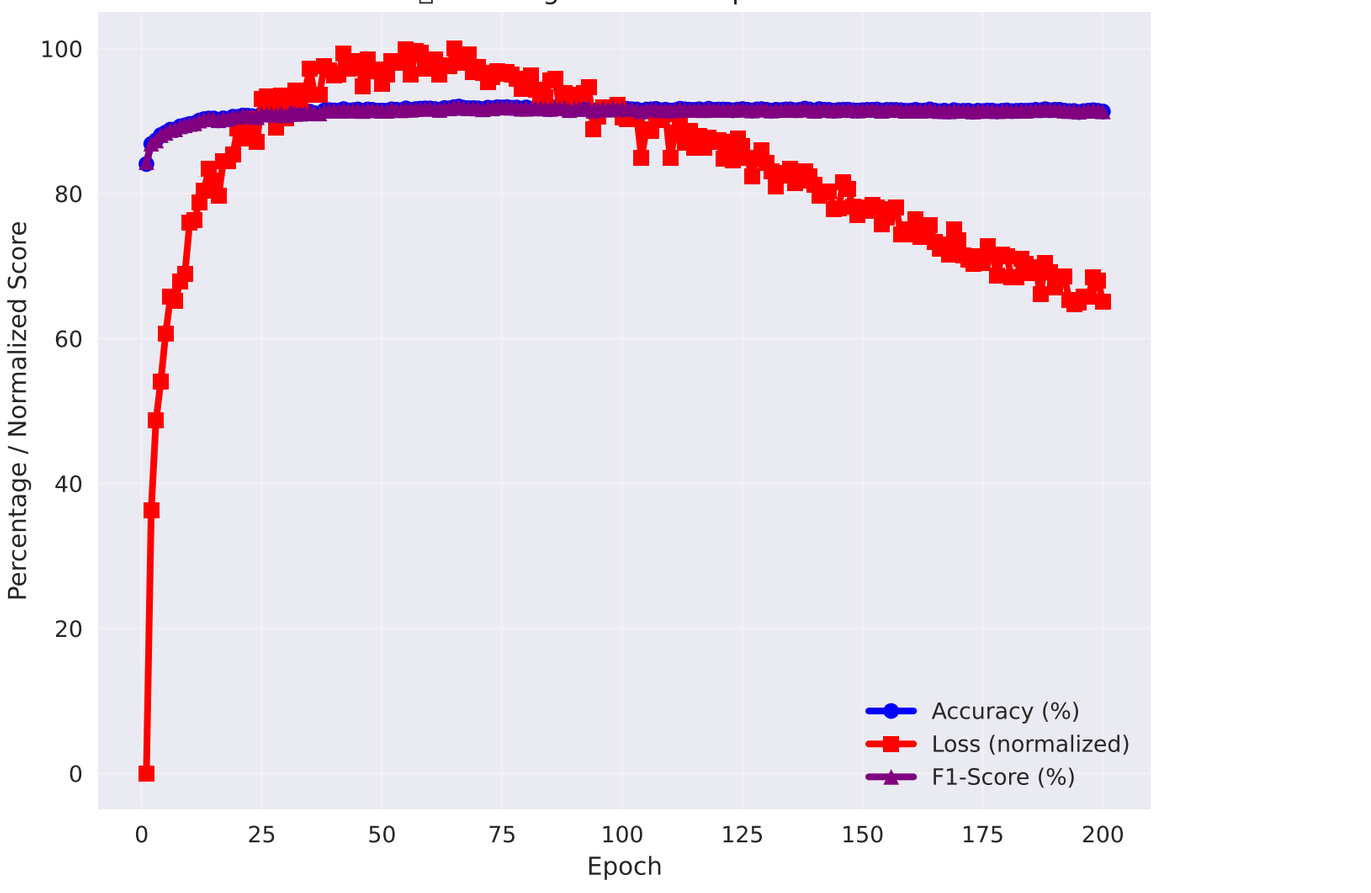


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-07-22 12:31:52
EPOCHE TOTALI: 200

METRICHE FINALI:
• Accuracy: 91.32%
• Loss: 3.1307
• Average Precision: 0.914
• Average Recall: 0.913
• Average F1-Score: 0.913

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +7.25%
• Riduzione Loss: +1.3542
• Stabilità Accuracy (CV): 0.0098

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.996)
• Peggiore Precision: Classe 6 (0.757)
• Migliore Recall: Classe 8 (0.983)
• Peggiore Recall: Classe 6 (0.751)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.850, R=0.859, F1=0.855
• Classe 1: P=0.996, R=0.974, F1=0.985
• Classe 2: P=0.868, R=0.872, F1=0.870
• Classe 3: P=0.916, R=0.909, F1=0.913
• Classe 4: P=0.855, R=0.871, F1=0.863
• Classe 5: P=0.981, R=0.981, F1=0.981
• Classe 6: P=0.757, R=0.751, F1=0.754
• Classe 7: P=0.945, R=0.978, F1=0.961
• Classe 8: P=0.987, R=0.983, F1=0.985
• Classe 9: P=0.980, R=0.954, F1=0.967

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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